Presentation Video — narration, v1

iGEM Claremont-CA 2026 | draft of 2026-10-24

Timecodes assume 140 words per minute; the uploaded file has a hard limit of 15:00.

Line numbers run continuously — send edits back as *line 212: …*

Opening

106 words ends 0:45

ON SCREEN — Team logo, a team photo, then the title card.

We're iGEM Claremont-CA, a first-year team in California. This is Phagenomics.

Our question is what decides which bacteria a phage can kill. We went at it four times this season. Three of those attempts failed, and the failures are the useful part — so that's how this video is organised.

Our project is about one question: what decides which bacteria a given phage can kill? We went after that question four times over the season. Three of those attempts failed, and the failures turned out to be the most useful thing we have — so this video is built around them rather than around a result.

Act I.1 extbar The bottleneck

70 words ends 1:15

ON SCREEN — Plaque assay animation; then the same drop on two other strains, no effect.

A drop of phage goes onto a lawn of bacteria, and a few hours later there's a clear circle where the bacteria used to be. Move the same drop to the next plate and nothing happens.

Same phage, different strain. Phages are fussy, and that fussiness is the bottleneck for using them as medicine: you need one that matches the strain in front of you, and usually there isn't one.

Act I.2 extbar Adsorption and the sugar coat

63 words

ends 1:42

ON SCREEN — Cross-section of a cell: membrane, wall, O-antigen, capsule. Mark which layer the phage meets first.

It gets decided at the first step, before any DNA moves: the phage has to recognise the outside of the cell and stick to it.

And many bacteria aren't presenting a bare protein surface. They're wrapped in sugar — long chains called the O-antigen, and above that, in some species, a capsule. The first thing a phage touches is a forest of sugar.

Act I.3 extbar Where the project came from

61 words

ends 2:08

ON SCREEN — Team photo with Prof. Libeskind-Hadas, captioned as an early conversation.

One thing belongs at the start, because it's why half this project exists.

Early on, Professor Ran Libeskind-Hadas at Claremont McKenna suggested reconstructing an ancestral phage — a whole organism, inferred from its relatives.

We couldn't build one. But scale that down to a single enzyme and you can order it from a synthesis company. That is what this project became.

Act I.4 extbar Our system, and the first clue

105 words

ends 2:53

ON SCREEN — Sampling at the California Botanic Garden, the genomes, then the three O-antigen cassettes side by side. Label the last one a hypothesis, not a prediction.

Our soil came from the California Botanic Garden in May. Out of it we got three distinct strains of *Pseudomonas* and a phage, Phi-8, that infects exactly one of them. The bacterial genomes are published; our team assembled and annotated the phage.

Then we lined the three strains up at the locus that builds their sugar coat. Same position, same flanking gene — and inside, three cassettes with no homology to each other.

Three strains, three different sugar coats, one phage that kills one of them. So our first hypothesis: the sugar coat might be one of the main things setting what Phi-8 can attack.

Cycle 1 extbar Design

74 words ends 3:25

ON SCREEN — DBTL bar with DESIGN lit. Lock and key.

But that isn't the hypothesis we tested first.

The textbook picture of attachment is a receptor-binding protein on the tail meeting a protein on the bacterial surface. Lock and key. If that's what you assume, the obvious move is to work out which binding protein fits which receptor.

If we'd known the receptor was a sugar we'd never have run this. We thought it was a protein, so we went looking for a protein.

Cycle 1 extbar Build

78 words ends 3:58

ON SCREEN — BUILD lit. Funnel from 441 to 178 to 114 $imes$ 63. Three tools and why each one stalled.

Four hundred and forty-one candidate receptors, clustered and narrowed to a hundred and fourteen binding proteins against sixty-three receptors — seven thousand, one hundred and eighty-two pairs, each folded as a complex on our university's GPU cluster and ranked by the interface score.

Getting it to run took three tools — ColabFold throttled by its alignment server, Boltz-2 blocked by the cluster's CUDA version — and we ended up on ESMFold because it was the one that ran.

Cycle 1 extbar Test

116 words ends 4:48

ON SCREEN — TEST lit. The ranking first, so it looks reasonable; then the control figure. Hold 0.137 and 0.136 large.

And we got a ranking. It looked fine — which is the trouble with this kind of output. It always looks fine.

What made us uneasy was that the scores tracked protein length more than anything else. So we ran the method against pairs where the answer was already known — complexes solved experimentally, against the same receptors shuffled into nonsense.

T5's pb-five with FhuA, a real co-crystallised complex, scored zero point one three seven. The same receptor shuffled scored zero point one three six.

We assumed that was the tool, so we rebuilt it with real evolutionary alignments. It got worse: a deliberately mismatched pair came out at zero point five, above every genuine binder.

Cycle 1 extbar Learn

220 words ends 6:22

ON SCREEN — LEARN lit. Two confidence bars: on scrambling the left collapses 0.55 to 0.17, the right does not move. Then two circles that never intersect.

Two tools, two control gates, both failed. One failure is about your run; two independent failures are about the method.

These models report two confidences: one for the shape of the molecule, one for how two molecules sit together. When we shuffled a sequence, the shape confidence collapsed from zero point five five to zero point one seven — the model could tell it was nonsense. But the interface number barely moved. It had never been reading an interface at all.

The reason is that these models don't simulate two proteins touching. They read evolution — looking across thousands of species for positions that change in step. If a residue on one side mutates and one on the other mutates to match, those two are probably in contact.

That only exists if the two proteins have been inherited together, in one genome, for a long time. A phage protein's relatives are in other phages; a bacterial protein's are in other bacteria. They're never in the same genome, so the record of that interface doesn't exist. Not hidden — never written.

So this was never a result about our proteins. It's a property of the

method for any phage-and-host pair, and more compute doesn't help. Ten control folds decided the fate of seven thousand, and we never ran the real screen.

Cycle 2 extbar Design and Build

49 words ends 6:43

ON SCREEN — PhageRBPdetect over 80 proteins, 8 candidates, rung-period diagram.

If we couldn't ask what the tail spike binds, we could ask what it is.

We ran an open-source classifier across all eighty Phi-8 proteins, folded the eight candidates, and classified the folds by the spacing of their backbone hydrogen bonds — which needs no structure database at all.

Cycle 2 extbar Test and Learn

150 words ends 7:48

ON SCREEN — Fold table, eight proteins and four verdicts — the method discriminates. Then the beta-helix render and a sugar-cutting animation.

Eight proteins, four different verdicts — so the method discriminates rather than agreeing with us. CDS 11, the tail spike, came back a right-handed beta-helix: a long triangular prism. And CDS 24, which the annotation had written off as hypothetical, came back as a second one.

That fold is a signature. Enzymes that chew through long sugar chains look like this, because a sugar chain is a thread and you need something that holds it straight while you cut. You can recognise a wood plane by its shape without being told what it cuts.

So the tail spike isn't a key. It's a blade — and the lock we'd spent months

looking for doesn't exist.

That reading arrived from two directions at once: our classifier flagged it as a depolymerase, and our PI reached the same conclusion from his own structural search. Between us we turned it into a hypothesis.

Cycle 2 extbar The hypothesis 68 words ends 8:17

ON SCREEN — One-sentence hypothesis box on white, held. Credited jointly with the PI.

In bacteria that wrap themselves in sugar, the ceiling on host range is set by what the depolymerase can digest — not by which protein the tail binds. And the narrow specificity we see now is a narrowed-down version of something broader further back.

That tells you what to measure — substrate range, not binding — and where to look for the broader version: back up the tree.

Cycle 3 extbar Design and Build 122 words ends 9:09

ON SCREEN — Shaft versus body rigidity, the six constructs, the tree collapsing to three nodes.

Two enzymes sit in the tail module: CDS 11, the beta-helix, and next to it CDS 12, a glycoside hydrolase. CDS 10 is still unknown, so we expressed that one too.

For the tail spike we took residues three-oh-five to nine-four-one, because the shaft drifts between models while the body stays rigid.

Then the ancestors — the part that came out of that first conversation.

We worked out the direction with our PI; he ran the tree and the reconstruction on sequences we prepared.

We picked three nodes. Six constructs in all, His-tagged, from Twist. The ancestors are exactly the same length as the wild type, which makes them indistinguishable on a gel — we should have thought about that before ordering.

Cycle 3 extbar Test

72 words ends 9:40

ON SCREEN — Four test rounds; the 9-29 gel; the 10-10 nickel column with blank elutions.

Four rounds at the bench. On September the twenty-ninth, CDS 12 gave an IPTG-dependent band at the right size while the tail spike gave nothing. On October the tenth we ran a nickel column on one ancestor, and both elutions came off blank — while the load and the flow-through carried the same band.

We read that as a purification we'd botched, and were about to spend another week on the column.

Cycle 3 extbar Learn: an outside view changed a result

207 words ends 11:09

ON SCREEN — The strongest human-practices beat in the video. Tolstorukov's email, then the 10-19 dot blot with 2.9$times$ marked.

What stopped us was a conversation.

On October the thirteenth we took it to Professor Ilya Tolstorukov at Keck

Graduate Institute, who works on protein expression. He wrote back before we met to say he wasn't sure he was the right person — expression is his field, purification isn't — and met us anyway. That turned out to be the point.

He changed the question. Not *why is nothing eluting*, but *is there anything there to elute*. Check whether the protein is being made at all — a dot blot against an anti-His antibody — before touching the column again.

Six days later we ran that blot. The flow-through from the ancestor read two point nine times the negative control.

So the protein was there. It had been made, it came off in the flow-through, and the run failed at capture — not at expression. A gel can't tell those apart; all three look identical. One cheap assay separates them, and it pointed the fix at the column instead of the construct.

We'd spent a week debugging the last step of a workflow without checking the first step worked, because the last step was the one visibly failing. That's the generalisable mistake, and one meeting fixed it.

Cycle 3 extbar What we still owe

31 words ends 11:22

ON SCREEN — Honest list. Caption: fifth consecutive assay with no empty-vector lane.

One thing we still owe: we've run five protein assays without an empty-vector BL21 lane, the control that separates our bands from host background. That one is on the bench now.

Cycle 4 extbar Design and Test

81 words ends 11:57

ON SCREEN — Colonies from the plaque zone, whole-genome sequencing, F126L inside a 30 kb capsular cluster.

Three cycles in, every guess about the receptor had come from the phage's side. So for the fourth we let the bacteria answer.

In July we picked colonies from inside the plaque zone — cells still growing where the phage should have killed them — and sequenced them against their parents.

Every difference was checked back against the raw reads. Exactly one survived: a single amino-acid change in a sugar phosphotransferase, inside a thirty-kilobase capsule cluster. The O-antigen cassette was untouched.

Cycle 4 extbar Learn

93 words ends 12:36

ON SCREEN — Diagram of a capsule hiding an intact O-antigen — how both conclusions hold.

That disagrees with Cycle 2, which is the whole reason for doing it this way round. Cycle 2 said the receptor is a sugar; Cycle 4 agrees it's a sugar and disagrees about which one. It points at the capsule, not the O-antigen underneath.

Both can be true at once, because a capsule can hide an intact O-antigen underneath it. We don't yet know which surface Phi-8 reads.

And the honest gap: those colonies were never confirmed resistant. They were colonies in a plaque zone, which is suggestive and isn't the same

thing.

Coda extbar The garden

167 words ends 13:48

ON SCREEN — Interview clips, at least 20 seconds. Credit both hosts and note permission.

Two days before our wiki froze we went back to the garden the soil came from, and asked whether a treatment as strain-specific as ours would be any use to them.

Most of what we'd assumed was wrong.

We expected to argue for biological control against a chemical default, and didn't have to — they already suppress soil fungi by mulching. We expected the objection to be ecological risk. It wasn't; it was whether the thing works. Copper is the only chemistry they have for bacterial disease, it goes onto every blossom every four or five days, and they have eighty-six acres and eight staff. The objection is arithmetic.

And the finding that costs us most: they've never had a sample come back positive for *Pseudomonas*. We isolated a soil phage and its environmental host — not a treatment for a disease this stakeholder has.

What they do need is something for bacteria inside the vascular system, where nothing they have reaches. Our strain doesn't solve that.

Coda extbar Safety

104 words ends 14:33

ON SCREEN — The IBC letter, 0 VFDB / 0 CARD, and the open condition quoted.

On safety: our biosafety committee gave conditional approval with one condition still open — and that condition is a question about host range, which is the subject of this entire project. We don't have full approval; the revision is being written.

Phi-8 carries no virulence factors and no resistance genes, and our constructs are protein fragments in lab *E. coli* — nothing here is a modified phage. But understanding what sets the ceiling on host range is the same as knowing what would have to change to raise it, so we treat this as dual-use rather than arguing our way to a clean no.

Coda extbar Where we are

147 words ends 15:36

ON SCREEN — Three columns fading in: Established / Predicted / Open.

So, where we actually are.

Established: the genomes, three unrelated sugar cassettes in one locus, a tail spike that turns out to be a sugar-cutting enzyme, a controlled negative result about interface prediction, and one mutation verified in the reads.

Predicted and partly tested: that Phi-8 won't infect the other two strains. A host-range matrix in July pointed the right way, but carried no no-phage control, so we won't call it confirmed.

Still open: clean expression, an activity assay, which sugar layer the phage reads, and whether Phi-8 is temperate.

Four cycles, three of them wrong, each wrong in a way that told us where

to go next. If there's one thing we'd pass on: put the cheap control in front of the expensive experiment. Ten folds told us what seven thousand couldn't, and one afternoon's dot blot told us what a week on a column hadn't.

2184 spoken words. With six act cards and a 30-second credit card, the finished file runs about 16:21 at 140 words per minute.